

CommitTx(*t*) $\triangleq$

$\wedge tx_status[t] = \text{"Active"}$

$\wedge pc[t] > Len(Workload[t].ops)$

$\wedge \text{LET } IsConcurrent(t_other) \triangleq$

$tx_status[t_other] = \text{"Committed"} \wedge$

$tx_commit_time[t_other] > tx_snapshot[t]$

HasConflict $\triangleq$

$\exists t_other \in AllTxns :$

$\wedge IsConcurrent(t_other)$

$\wedge \exists k \in tx_checkset[t] :$

$\exists v \in store[k] :$

$v.writer = t_other \wedge v.cts = tx_commit_time[t_other]$

IN IF *HasConflict*

THEN $\wedge tx_status' = [tx_status \text{ EXCEPT } ![t] = \text{"Aborted"}]$

$\wedge \text{UNCHANGED } \langle client_dep, client_last_commit, shard_ct,$
 $tx_snapshot, tx_checkset, tx_buffer,$
 $tx_commit_time, pc, store, op_seq \rangle$

ELSE $\wedge shard_ct' = shard_ct + 1$

$\wedge tx_commit_time' = [tx_commit_time \text{ EXCEPT } ![t] = shard_ct']$

$\wedge tx_status' = [tx_status \text{ EXCEPT } ![t] = \text{"Committed"}]$

$\wedge store' =$

$[k \in Keys \mapsto$

IF $tx_buffer[t][k] \neq \langle \rangle$

THEN $store[k] \cup$

$\{[val \mapsto tx_buffer[t][k][1],$

$cts \mapsto shard_ct',$

$writer \mapsto t]\}$

ELSE $store[k]$

$\wedge client_dep' =$

IF $shard_ct' > client_dep[Workload[t].session]$

THEN $[client_dep \text{ EXCEPT } ![Workload[t].session] = shard_ct']$

ELSE $client_dep$

$\wedge client_last_commit' =$

$[client_last_commit \text{ EXCEPT}$

$![Workload[t].session] = shard_ct']$

$\wedge \text{UNCHANGED } \langle tx_snapshot, tx_checkset, tx_buffer, pc, op_seq \rangle$